

Faculty of Statistical and Computer Engineering
Universidad Nacional del Altiplano (UNAP)
Av. Floral 1153, Puno 21001, Peru
75521963@est.unap.edu.pe

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The Editor-in-Chief

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Dear Editor,

We submit for your consideration the manuscript “*Harmonization and validation limits of satellite Secchi retrieval in Lake Titicaca*” as a research article.

Lake Titicaca (8,372 km², 3,810 m a.s.l.) is the largest lake in South America and, by volume, the largest tropical lake in the world. It is a transboundary Ramsar site and the freshwater reserve for millions of Andean people, yet its water quality is documented only by infrequent field campaigns. It is precisely the kind of large lake, outside the well-instrumented systems, that we understand the journal’s scope to cover.

The manuscript is as much about *how to evaluate* a machine-learning water-quality product as about the product itself. We assemble 1,002 real satellite–field match-ups (812 carrying a Secchi reading) against twelve years of IMARPE limnological monitoring, and show that two defaults imported from the turbid-water literature fail in clear, campaign-sampled water. For each we give a diagnostic that any study can run.

(1) Land-derived cross-sensor harmonization is invalid over dark water. The coefficients of Roy et al. (2016), fitted on vegetation and soil and widely reused to merge Landsat into Sentinel-2 archives, carry additive intercepts that exceed the entire native over-water Landsat signal by factors of 1.4 to 69.8. A classifier still identifies the source sensor from the *already harmonized* reflectance with 99.8% accuracy against a 66.3% chance rate: the retrieval was learning a sensor label, not unified optics. We propose a classifier two-sample test as a one-line check for this failure.

(2) The blocking unit for cross-validation must be diagnosed, not assumed. Blocking by monitoring station is the standard remedy for spatial autocorrelation, but in a campaign-based programme it leaves the dominant dependence untouched: under station-blocked folds a null model that knows only the campaign date still reaches $R^2 = 0.305$, as much as a zone-mean null reaches under any design. Station blocking leaves the reported accuracy unchanged from a random split ($R^2 = 0.596$ in both cases), which is evidence that it retains nothing rather than that no leakage exists. We give the null-model procedure — three nulls under each candidate blocking unit — that identifies the unit carrying the dependence.

(3) An honest retrieval with calibrated uncertainty. Under campaign-blocked validation a Random Forest retrieves Z_{SD} with $R^2 = 0.574$ and RMSE 1.86 m ($n = 812$), against 0.097 for a correctly bounded classical blue/green band ratio. Split-conformal intervals are well calibrated marginally (89.4% empirical coverage at a 90% nominal level) but not conditionally, falling to 71.3% in the clearest water — a limitation a single coverage figure would hide. Skill collapses under

leave-one-zone-out extrapolation, and we report explicitly that chlorophyll-*a*, suspended solids and thermal temperature are *not* retrievable in this oligotrophic regime.

We are aware that our headline accuracy is modest by the standards of the turbid-lake literature. That is the point: we show it is the honest number for this class of water under a defensible design, and that the higher figures such studies report are partly an artefact of the blocking unit and of baselines implemented as straw men.

All data and code are openly available at <https://github.com/Andre031222/harmonized-titicaca-transparent>. The analysis is reproducible end to end from the deposited match-up tables without Earth Engine credentials, and every number, table and figure in the manuscript is generated by that pipeline rather than typed by hand.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission and declare no competing interests. We have no preferred or excluded reviewers.

Thank you for considering our work.

Sincerely,

Richar Andre Vilca-Solorzano
(on behalf of all authors)